

Surveillance / SLAM / Video Streaming

Supplementary Design Document

Sub-document to Unified SDD v1.5

문서번호 **SAN-SDD-SUR-001**

v1.5

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Document ID	SAN-SDD-SUR-001
Version	v1.5
Classification	Supplementary SDD (sub to Unified SDD)
Preceding Docs	SAN-SDD-SWARM-001 v1.5, SAN-IDS-CMD-001 v1.5

Revision History

Version	Changes
v1.0	Initial release. 4 design changes specified.
v1.1	Cover classification table updated (author/schedule removed).
v1.2	PM decisions D1~D7 + A1~A5 applied: (D1) Squadron Follower loss (8 -> 7) (Leader S1 + Hub UGV S2 + Followers S3-S8 = 6 followers); (D2) Leader succession deputy_chain = [2, 3, 4, 5, 6, 7, 8], Hub UGV(S2) is priority 1; (D3) CycloneDDS NIC binding specified; (D7) AI inference Phase 1 RK3588 / PoC Hailo M.2; Section 3 sector distribution table updated for 8 robots, Section 4 Hub UGV S2 specified.
v1.3	Cost Map + SLAM 5s + LTE Redundancy: (1) §5 SLAM aggregation period **5s** (v1.3) -> **5 sec** ; (2) §4 Robosense E1 VFOV **25° -> 90°** correction; (3) UGV spec table (1300x850mm, 12km/h, slope 30°, obstacle 235mm, ditch 220mm); (4) LTE redundancy — Follower S3 LTE backup activates on Hub LTE failure.
v1.4	Deputy UGV pre-designation reflected: (1) Section 5 SLAM aggregation — Deputy UGV (S3) takes over on Hub UGV failure; (2) Deputy UGV spec is identical to Hub UGV (dual SBC + LTE).
v1.5	DCN-2026-001 applied (2026-05-12): (1) Sub-document reference updated to SAN-SDD-SWARM-001 v1.5; (2) §3.2 added explanatory note for Deputy UGV (S3) shadow surveillance behavior (covers rear 180° as Hub-takeover standby); (3) Body "9-robot" / "9 robots" references corrected to "8-robot" / "8 robots" (SAN_SwarmOperation_Integrated v1.0 deprecation residual); (4) Related Docs version pins v1.4 / v1.1 → v1.5 throughout.

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1. Overview

1.1 Document Position

This Supplementary SDD is a sub-document of SAN-SDD-SWARM-001 v1.5 (Unified SDD), specifying the following four design decisions formally:

1. 8-robot 360° surveillance distribution — Leader assigns each follower a front-180° area and Hub UGV takes the rear 180°
2. Follower Pan-tilt low-speed sweep during driving — active 360° coverage while moving
3. Hub UGV dual SBC architecture — split between SLAM fusion SBC and Comm/Video SBC
4. SLAM collection period change (1 Hz → ****5s**** (v1.3)) and video streaming GStreamer UDP/SRT adoption

1.2 Change Summary

#	Topic	Previous (v1.0)	This Document
S1	Surveillance distribution	Not specified (forward-only recon)	Dynamic Leader-driven 360° share (10s)
S2	Follower Pan-tilt	Only when tracking target	Slow sweep while driving (30°/sec)
S3	Hub UGV SBC	Single RK3588J	Dual SBC (SLAM + Comm/Video split)
S4	SLAM collection period	Follower -> Hub 1 Hz	Aggregated bundle every **5s** (v1.3)
S5	Video streaming	Protocol unspecified	GStreamer UDP/SRT, Android pull mode

1.3 KPP Impact

- ****Real-time video $\leq 200\text{ms}$ **** — GStreamer SRT live latency=120ms tune achieves target
- ****Target classification $\geq 90\%$ **** — Active Pan-tilt sweep minimizes blind spots
- ****Swarm comm latency $\leq 150\text{ms}$ **** — Relaxed SLAM period frees bandwidth headroom

2. Camera Surveillance Area Definition

2.1 Camera HFOV Specifications

Mode	Use	Zoom	HFOV (Day)	HFOV (Thermal)
Surveillance	Wide sweep	1x (Wide)	~60°	~24°
Observation	POI track	3-5x	~12-20°	~5-8°
Identification	Target ID	10-15x	~4-6°	~1.5-2.5°
Engagement	Precision aim	20-30x	~2-3°	~0.8-1.2°

Unless otherwise specified, Surveillance Mode (Wide, HFOV 60°) is the default. For wide-area surveillance, 60° HFOV provides $\geq 30^\circ$ overlap between adjacent followers.

2.2 Leader (Go2) Fixed Surveillance

- Direction: Forward 60° (heading-aligned)
- Range: $\pm 30^\circ$ (30° each side of heading)
- Zoom: 1x Wide (HFOV 60°) fixed
- Data: Real-time video to Hub UGV (optional broadcast)

The Leader's surveillance area auto-tracks heading. Leader focuses on 1s prediction computation and recon, so low camera autonomy simplifies system complexity.

2.3 Follower Pan-tilt Dynamic Surveillance

2.3.1 Pan-tilt Operation Modes

Mode	Pan Motion	Tilt Motion	Transition Trigger
Sweep	Reciprocating in sector	Horizon-based -5° to +10°	Default while driving
Fixed	Hold commanded direction	Hold commanded angle	Leader command or operator
Track	Auto-follow target	Auto-follow target	AI detection or threat
Engage	Target lock-on + lead	Target lock-on + lead	Just before Confirm Fire

2.3.2 Sweep Rotation Speed

$\omega_{\text{sweep}} = (\text{sector_width} - \text{HFOV}) \times 2 / \text{period}$ Example: sector_width = 90°, HFOV = 60°, period = 6s
 $\omega_{\text{sweep}} = (90 - 60) \times 2 / 6 = 10^\circ/\text{sec}$
Example: sector_width = 180°, HFOV = 60°, period = 8s
 $\omega_{\text{sweep}} = (180 - 60) \times 2 / 8 = 30^\circ/\text{sec}$

Standard sweep speed 30°/sec. On threat detection, immediate switch to Track mode for target lock-on. Night thermal mode uses half speed (15°/sec) due to narrower HFOV (~24°).

2.3.3 Pan-tilt Stabilization During Driving

Vehicle vibration on pan-tilt video is absorbed by gimbal stabilization. The RCWS gimbal includes active IMU-based stabilization meeting the 0.05° tracking error KPP. Sweep mode is area scanning (not target tracking), tolerating wider vibration than Track mode.

2.4 Hub UGV Surveillance Area

- Pan range: heading +90° to +270° (rear 180°)
- Sweep speed: 30°/sec (8s period)
- Threat detected: immediate Track mode + position broadcast to nearby followers

3. Surveillance Area Distribution Policy

3.1 9-Robot 360° Distribution Principles

Robot	ID	Direction	Primary Mode	Role
Leader (Go2)	S1	Front 0° ±30°	Fixed (front)	Recon + path clearance
Hub UGV	S2	Rear ±90° (180°)	Sweep 30°/sec	Primary rear surveillance + comm/SLAM hub + Leader succession priority 1
F1 (Follower)	S3	Front-left -80° to -30°	Sweep	Front-left surveillance
F2 (Follower)	S4	Front-right +30° to +80°	Sweep	Front-right surveillance
F3 (Follower)	S5	Left flank -130° to -80°	Sweep	Left flank
F4 (Follower)	S6	Right flank +80° to +130°	Sweep	Right flank
F5 (Follower)	S7	Left rear -180° to -130°	Sweep	Left rear flank
F6 (Follower)	S8	Right rear +130° to +180°	Sweep	Right rear flank

3.2 Front 180° Distribution Algorithm

Robot	ID	Sector	Center	Primary Direction
Leader	S1	-30° to +30°	0° (front)	Front 60°
Hub UGV	S2	+90° to -90° (wrap)	180° (rear)	Rear 180° sweep
F1	S3	-80° to -30°	-55° (front-left)	Front-left
F2	S4	+30° to +80°	+55° (front-right)	Front-right
F3	S5	-130° to -80°	-105° (left flank)	Left flank
F4	S6	+80° to +130°	+105° (right flank)	Right flank

Robot	ID	Sector	Center	Primary Direction
F5	S7	-180° to -130°	-155° (left rear)	Left rear flank
F6	S8	+130° to +180°	+155° (right rear)	Right rear flank

Each follower's assigned sector is 60° wide and camera HFOV is 60°, so sweep can degenerate to fixed gaze. For active gimbal operation, sectors are usually allocated slightly wider (80-90°) with 30°/sec sweep.

★ **v1.5 Deputy UGV (S3) Shadow Coverage Note (DCN-2026-001 D-001):**

Deputy UGV (S3) is positioned in the same column as Hub UGV (S2) with a small longitudinal offset (~3 m behind Hub). S3 operates the same camera/Pan-tilt and Robosense E1 LiDAR as Hub but its surveillance sector is also Rear 180° in "shadow mode": S3 covers the same arc as Hub UGV but with phase-offset sweep (starts at +90° when Hub is at -90°) to maximize revisit rate of the rear arc. On Hub failure, Deputy seamlessly assumes the full Rear 180° sector with no re-tasking delay (validated in TST S18-3, ≤ 7 s full takeover).

3.3 Hub UGV Rear 180° Coverage

- Pan range: chassis heading +90° to -90° (rear hemisphere)
- Sweep speed: 30°/sec
- Sweep period: ~8s (180° + 60° HFOV round trip)
- Threat detected: Track mode + cross-cue with F6, F7

Rear surveillance counters ambush and flanking infiltration. Combination of Hub's central-rear position and 360° pan-tilt capability ensures rear security.

3.4 Periodic Refresh

Trigger	Refresh Period	Refresh Scope
Periodic refresh	10s	Recompute all follower sectors
Formation change	Immediate (post-transition)	Full reassign for new formation
Follower lost	Immediate (post-reform)	Redistribute 360° among remaining
Threat detected	Immediate (while present)	Priority assign to threat-side followers
Manual command	Operator immediate	Concentrate/disperse over specific area

10s period rationale: (1) Longer than sweep period (~8s), avoiding mid-sweep sector changes, (2) Fast enough for formation drift, (3) Negligible comm load (35 B/s for 7 followers). Adjustable 5-30s by PM.

3.5 Dynamic Adaptation

3.5.1 Threat Sector Focus

When RF scanner or any follower detects a threat, Leader reassigns 2-3 closest followers to the same threat sector. Yields (1) higher detection probability, (2) better cross-cueing accuracy, (3) faster engagement.

3.5.2 Gap Filling

On follower loss (comm loss, power loss), remaining followers redistribute sectors automatically. 9 → 8 transition increases average sector width $30^\circ \rightarrow 35.7^\circ$, full coverage guaranteed.

3.5.3 Recon vs Defense Mode

Offensive mode (assault): narrow front sectors ($15\text{-}20^\circ$ /follower), wide rear. Defense mode (point defense): balanced (45° /follower) front-rear.

4. Hub UGV Dual SBC Architecture

4.1 Dual SBC Rationale

Hub UGV (robot_id = S2) used a single RK3588J SBC in v1.0. The following v1.1 concurrent requirements clearly exceed single-SBC capacity:

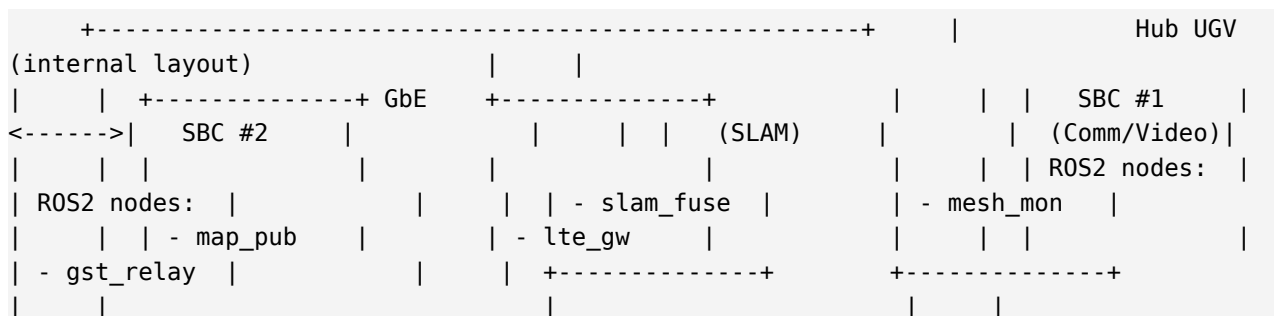
- 8-robot SLAM data fusion (multirobot_map_merge + g2o)
- Wi-Fi 6 Mesh routing oversight (OpenWrt batman-adv)
- Military LTE gateway (mwan3)
- Video stream forwarding (GStreamer SRT/UDP)
- DDS Security encrypt/decrypt (AES-256-GCM)
- 360° surveillance dispatch

→ RK3588J would exceed 90% utilization causing latency spike, packet drop, SLAM delays. Solution: deploy two SBCs to split workload.

4.2 Dual SBC Role Split

Aspect	SBC #1 (SLAM)	SBC #2 (Comm/Video)
Model	RK3588J or Jetson Orin	RK3588J
Memory	16 GB LPDDR5	8 GB LPDDR5
Storage	256 GB NVMe	128 GB NVMe
Primary task	SLAM aggregated fusion	Wi-Fi6 Mesh + LTE gateway
Secondary task	Global map broadcast	GStreamer SRT/UDP forwarding
Processes	slam_toolbox, multirobot_map_merge	DDS, GStreamer, OpenVPN/IPsec
Network	Internal GbE (direct to SBC#2)	Wi-Fi 6 (ext) + LTE modem + Internal GbE
Power	Main battery 12V rail	Main battery 12V rail
Cooling	Independent heatsink + fan	Independent heatsink + fan

4.3 SBC Interconnect Bus



v | | Wi-Fi 6 Mesh --- LTE --- Other Robots / Operator
| +-----+

4.4 SBC Failure Response

Failure	Behavior	Mission Impact
SBC #1 (SLAM) failed	Aggregated SLAM suspended, each follower runs local SLAM	Mission continues (no global map)
SBC #2 (Comm) failed	Other Wi-Fi6 mesh robots assist routing	Mission pause or SBC reboot
Both SBCs nominal	All functions	Full

5. SLAM Aggregated Collection

5.1 Period Change Rationale

Aspect	v1.0 (1 Hz)	v1.1 (5s aggregated (v1.3))
Comm load (8 robots)	90-360 KB/s	1-3 KB/s (1/100 reduction)
Hub SBC #1 load	Continuous 100% busy	Periodic 10-20% busy
Global map refresh	1s (excessive)	**5s** (v1.3) (sufficient)
Real-time avoidance	Global map dependent	Local SLAM/Nav2 (proper)
Bandwidth headroom	Low	Adequate for 1s prediction

5.2 SLAM Data Flow

[Each Follower] Local SLAM (slam_toolbox, 1Hz local)		Every **5s** (v1.3) send aggregated occupancy grid delta (SLAMLocalDelta)	v
[Hub SBC #1] multirobot_map_merge		- feature-based merge (ORB/RANSAC)	
- pose-graph optimization (g2o)	v	[Hub SBC #1] maintains global merged map	
		Every **5s** (v1.3) broadcast (SLAMAggregatedMap, or delta only)	v
		[All Robots] Nav2 cost map static layer update	

5.3 Dynamic Period Adjustment

Operational Situation	Recommended	Rationale
Wide-area recon	60s	Slow-changing terrain
General (default)	30s	Balanced
Urban/indoor	15s	Complex terrain
After formation change	Immediate (1x)	Re-align with new formation
Heavy obstacles	10s temp	Safety

5.4 Comm Load Verification

Per aggregated bundle in 8-robot squadron:

- Occupancy grid (100×100m, 0.1m resolution) = 1000 × 1000 cells = ~1 MB raw
- After PNG compression: ~50-200 KB
- 8 robots concurrent 30s: avg 30 KB/s, worst 60 KB/s
- Wi-Fi 6 Mesh capacity (12.5-37.5 MB/s) used < 1%

5.5 Local vs Global SLAM Role Separation

Role	Local (each robot)	Global (Hub aggregated)
Update rate	1 Hz	**5s** (v1.3)
Purpose	Real-time avoidance, Nav2	Long-term recon, global planner
Coverage	~30m around robot	Hundreds of meters - km
Storage	Each robot	Hub UGV SBC #1
Loss impact	Reduced avoidance capability	Mission planning info loss

6. Video Streaming — GStreamer UDP/SRT

6.1 Background

v1.0 only specified "LTE backhaul, BEST_EFFORT". v1.1 confirms:

- Encoder: H.265 (default) / H.264 (compatibility)
- Container: MPEG-TS
- Protocol: SRT (operator) / UDP (inter-robot)
- Pipeline: GStreamer
- Trigger: Android APP request (Pull mode)

6.2 SRT vs UDP Selection

Aspect	UDP (Plain)	SRT (adopted, operator)
Latency	Lowest (~50ms)	200-500ms (buffer)
Packet loss	None (glitches)	Auto-retransmit (ARQ)
Encryption	Separate impl	Built-in AES-128/256
Bandwidth adapt	None	Dynamic bitrate
Firewall traversal	Difficult	Easy (1 port)
Recommended	Inter-robot light	Operator screen
KPP 200ms met	Met	Met with live latency=120ms

6.3 GStreamer Pipelines

6.3.1 Sender (Follower or Hub) — H.265 + SRT

```
gst-launch-1.0 -v \          v4l2src device=/dev/video0 \          ! video/x-raw,width=1920,height=1080,framerate=30/1 \          ! x265enc bitrate=2000 tune=zerolatency speed-preset=ultrafast \          ! h265parse \          ! mpegtsmux \          ! srtsink uri="srt://<operator_ip>:8888?mode=caller&latency=120&passphrase=..."
```

6.3.2 UDP Variant (inter-robot)

```
gst-launch-1.0 -v \          v4l2src device=/dev/video0 \          ! video/x-raw,width=640,height=480,framerate=15/1 \          ! x264enc bitrate=500 tune=zerolatency \          ! h264parse \          ! mpegtsmux \          ! udpsink host=<hub_ip> port=5000
```

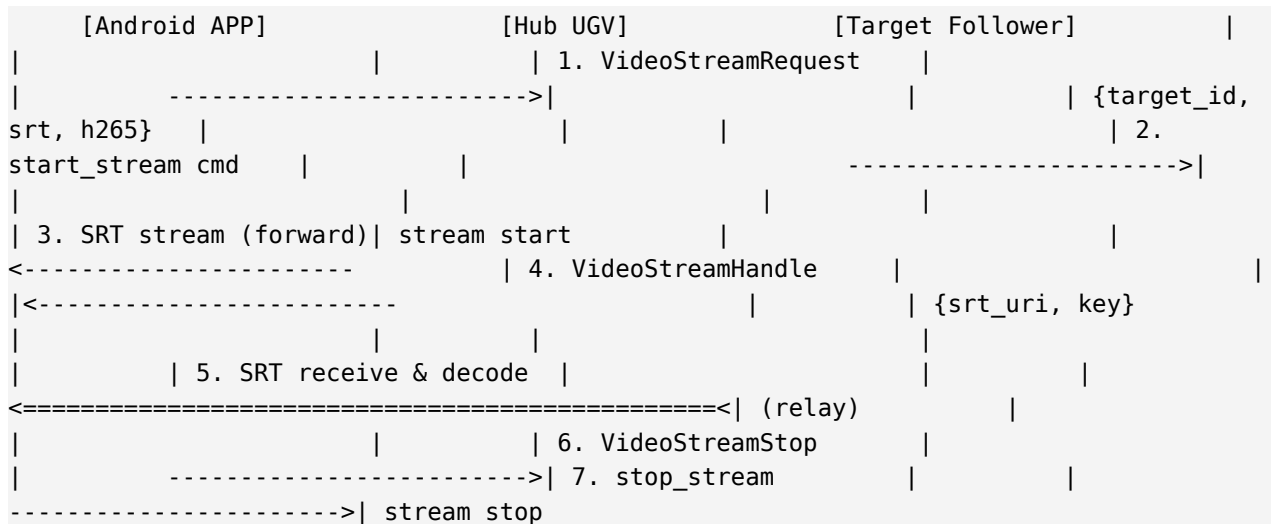
6.3.3 Hub Relay (Forwarding)

```
gst-launch-1.0 -v \          udpsrc port=5000 caps="application/x-rtp" \          ! rtpmp2tdepay \          ! tsdemux \          ! srtsink uri="srt://<operator_ip>:8888?mode=caller&latency=120"
```

6.3.4 Receiver (Android APP)

```
// Android (ExoPlayer)    val srtUri = Uri.parse("srt://<robot_ip>:8888?
mode=listener&latency=120")    val source =
ProgressiveMediaSource.Factory(dataSourceFactory)    .createMediaSource(MediaItem.
fromUri(srtUri))    exoPlayer.setMediaSource(source)    exoPlayer.prepare()
exoPlayer.playWhenReady = true
```

6.4 Video Request Flow (Pull mode)



6.5 Bitrate and Quality Policy

Quality	Resolution	FPS	Bitrate	Use
HD	1920x1080	30 fps	3-5 Mbps	Detailed recon, target ID
FHD	1280x720	30 fps	1.5-2 Mbps	General (default)
Low	640x480	15 fps	300-500 Kbps	Comm-constrained
Thumbnail	320x240	5 fps	100 Kbps	Multi-screen monitoring

6.6 Security

- SRT built-in AES-128 (performance-balanced)
- Passphrase: per-mission auto-issuance (DDS Security key mgmt)
- Hub-Operator segment: IPsec or OpenVPN tunnel
- Robot-Hub segment (LAN): WPA3 + DDS Security

7. Additional Message Definitions

Messages defined in this SDD are formally integrated in SAN-IDS-CMD-001 v1.1. This section summarizes.

7.1 SurveillanceSectorAssignment (new)

```
message SurveillanceSectorAssignment {          uint32  sequence          = 1;
uint32  robot_id          = 2;          float   sector_start_deg   = 3;
float   sector_end_deg    = 4;          uint32  valid_period_sec   = 5;
string  priority          = 6;          string  mode_hint          = 7;
uint64  timestamp_ms      = 8;          }          // QoS: P1, RELIABLE, 10s period + event
```

7.2 PanTiltCommand (new)

```
message PanTiltCommand {          uint32  robot_id          = 1;          float
target_pan_deg    = 2;          float   target_tilt_deg    = 3;          float
speed_dps         = 4;          string   mode              = 5;          float
sweep_range_deg   = 6;          uint64  timestamp_ms      = 7;          }          // QoS: P1,
RELIABLE
```

7.3 SLAMAggregatedMap (new)

```
message SLAMAggregatedMap {          bytes   occupancy_grid_png = 1;
Pose2D  origin      = 2;          float   resolution_m      = 3;
uint32  width_cells  = 4;          uint32  height_cells     = 5;
uint32  contributing_robots = 6;          uint64  timestamp_ms      = 7;          }          //
QoS: P2, RELIABLE, TRANSIENT_LOCAL, **5s** (v1.3)
```

7.4 SLAMLocalDelta (modified — replaces SLAMDelta)

```
message SLAMLocalDelta {          string   robot_id          = 1;
bytes   occupancy_grid_delta_png = 2;          Pose2D  origin              = 3;
float   resolution_m              = 4;          uint64  coverage_start_ms   = 5;
uint64  coverage_end_ms           = 6;          uint64  timestamp_ms        = 7;
}          // QoS: P2, RELIABLE, **5s** (v1.3)
```

7.5 VideoStreamRequest (new)

```
message VideoStreamRequest {          uint32  target_robot_id = 1;          string
protocol          = 2;          string   codec              = 3;          string   quality
= 4;          bool    encryption          = 5;          string   action          = 6;
uint64  timestamp_ms = 7;          }          // QoS: P1, RELIABLE
```

7.6 VideoStreamHandle (modified)

```
message VideoStreamHandle {          uint32  target_robot_id = 1;
string  srt_uri      = 2;          string   codec              = 3;
string  quality       = 4;          uint32  actual_bitrate_kbps = 5;
string  passphrase    = 6;          string   status              = 7;
uint64  timestamp_ms  = 8;          }          // QoS: P1, RELIABLE
```

8. Existing SDD Impact

Unified SDD §	Existing Spec	Impact	Action
§3 (Hub platform)	Single SBC	Changed to dual SBC	v1.1 absorbed
§9 (SLAM/comm)	Follower -> Hub 1 Hz	5s aggregated (v1.3)	v1.1 absorbed
§14 (Message)	31 messages	6 messages affected	IDS v1.1 absorbed

8.1 TST Test Scenario Impact

New ID	Verification Target	Test Environment
S15-1	8-robot 360° surveillance accuracy	Gazebo W5
S15-2	Pan-tilt sweep detection rate	Gazebo W2
S15-3	Hub SBC #1/#2 partial operation	Gazebo W1
S15-4	SLAM 30/60s global map consistency	Gazebo W3
S15-5	GStreamer SRT 200ms latency KPP	Real robot (LTE)
S15-6	Multi-follower simultaneous video	Real robot (operator APP)

9. Conclusion

This Supplementary SDD formally specifies the following four design decisions made in the SAN-SDD-SWARM-001 operational phase:

5. 8-robot 360° surveillance distribution (Leader 10s period, Follower front 180°, Hub rear 180°)
6. Follower Pan-tilt dynamic sweep (30°/sec during driving)
7. Hub UGV dual SBC (SLAM-dedicated + Comm/Video-dedicated)
8. SLAM **5s** (v1.3) collection + GStreamer SRT video + Android pull

The design improves operability and performance without changing v1.0 essence. Relaxing SLAM period frees bandwidth for 1s-prediction broadcast and video streaming.